

Ladle TAT Optimisation

Integrated Thermal, Kinetic and Scheduling Framework

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Problem Statement

Achieve Ladle TAT of 30–40 minutes across all IF+LF configurations while satisfying metallurgical, thermal and scheduling constraints.

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Abstract

Amalgam Steel’s current LF cycle of ~ 70 min far exceeds the 30–40 min TAT target. This report presents a four-module mathematical model that finds the *minimum feasible LF service time* satisfying all metallurgical and scheduling constraints simultaneously. The central finding is that the current LF time is not driven by physics — it is driven by sub-optimal slag chemistry. With enhanced dephosphorisation kinetics ($4.4\times$ baseline), 40 min IF pre-treatment, and proper ladle scheduling, a TAT of ≤ 40 min is achievable for 8 IF + 2 LF (Case 1), with LF arc energy of ≈ 27 kWh/T against a 100 kWh/T limit. A 2000-heat Monte Carlo simulation confirms $\geq 98\%$ of heats meet the TAT constraint under realistic stochastic variability.

All parameters marked as *Company Confirmed* are fixed inputs from Amalgam Steel.
Results are reproducible using the attached Python implementation (`ladle_tat_v5_final.py`).

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1 Problem Statement & Scope

Amalgam Steel operates 8 Induction Furnaces (IF) and 2 Ladle Furnaces (LF) on an IF → LF → Caster route producing mild steel grades E250, E350, and V55. Two linked problems motivate this study.

Heat sequencing: Mismatched IF-to-LF arrival rates cause queue build-up and ladle holds exceeding 15 min, chilling the steel below the casting threshold.

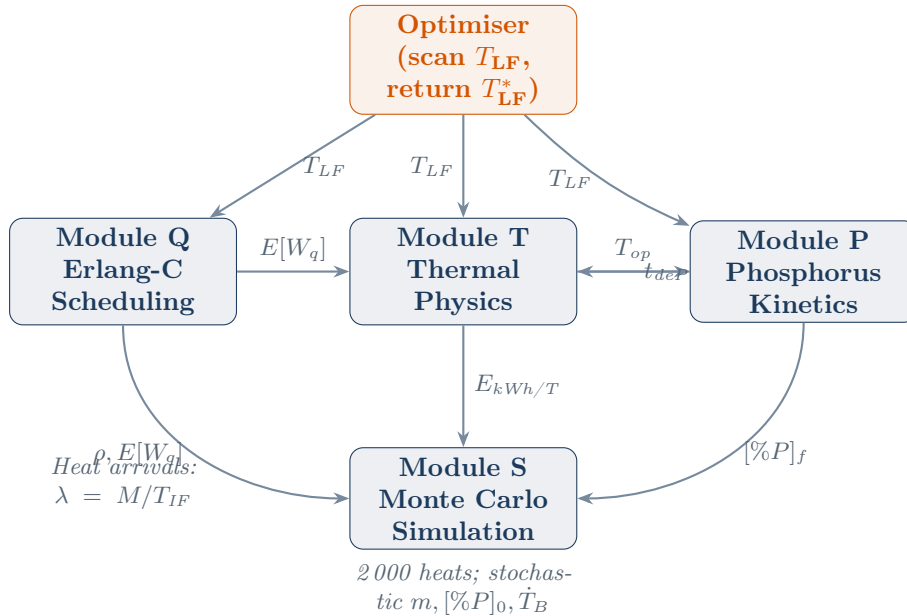
TAT exceedance: The current LF cycle (~ 70 min) far exceeds the target. Fixed process overhead (tapping, transport) consumes 25 min of the 40 min budget, leaving at most 15 min for queue wait *plus* LF service.

Constraint	Limit	Physical driver
Max ladle TAT	≤ 40 min	Caster sequence timing; steel temperature floor
Max queue hold ($E[W_q]$)	≤ 15 min	Radiative heat loss during waiting
Final phosphorus $[\%P]_f$	$< 0.040\%$	Grade spec; slab quality
LF arc energy E_{LF}	< 100 kWh/T	Transformer/utility limit
Casting temperature T_{cast}	≥ 1625 °C	Solidification; minimum superheat

The model is developed for three cases: (1) 8 IF + 2 LF, (2) 4 IF + 1 LF, and (3) the general M IF + N LF configuration.

2 Model Architecture

The optimisation suite comprises four tightly coupled modules. The coupling structure is not sequential — each module feeds data into the others:



Module Q sets queue stability and determines $E[W_q]$; Module T uses $E[W_q]$ as the Zone-B wait time to back-solve $T_{exit,LF}$ and compute arc energy; Module P uses the actual LF operating temperature from Module T to compute the temperature-corrected k_{LF} ; Module S drives all three stochastically.

3 Why Two-Stage Dephosphorisation is Mathematically Necessary

This section provides a rigorous proof that single-stage LF-only dephosphorisation is *physically impossible* within the 12–13 min LF window, regardless of slag enhancement. Two-stage processing (40 min IF pre-treatment + enhanced LF finishing) is therefore not a design preference — it is a mathematical necessity.

3.1 First-Order Kinetic Model

Dephosphorisation obeys first-order kinetics:

$$\frac{d[\%P]}{dt} = -k_{\text{LF}} [\%P] \quad \implies \quad [\%P](t) = [\%P]_0 e^{-k_{\text{LF}} t} \quad (1)$$

Inverting for the required rate constant given a time budget T_{LF} :

$$k_{\text{LF}}^{\text{required}} = \frac{\ln([\%P]_0 / [\%P]_{\text{target}})}{T_{\text{LF}}} \quad (2)$$

3.2 Maximum Achievable Rate Constant

The baseline rate constant $k_{\text{base}} = 0.0116 \text{ min}^{-1}$ is calibrated to the current practice (70 min LF $\rightarrow$ 0.04% P). The maximum enhancement via improved slag chemistry and Ar stirring is:

$$k_{\text{LF}}^{\text{enh}} = k_{\text{base}} \underbrace{\left(\frac{B_{\text{new}}}{B_{\text{old}}}\right)^{0.5}}_{f_B} \underbrace{\left(\frac{Q_{\text{new}}}{Q_{\text{old}}}\right)^{0.4}}_{f_Q} \underbrace{\left(\frac{[\%FeO]_{\text{new}}}{[\%FeO]_{\text{old}}}\right)^{0.5}}_{f_{\text{FeO}}} \quad (3)$$

Parameter	Baseline	Proposed	Factor	Formula
Basicity $B = \text{CaO}/\text{SiO}_2$	2.5	4.5	$f_B = 1.342$	$(4.5/2.5)^{0.5}$
Ar stirring Q (L/min/T)	0.10	0.40	$f_Q = 1.741$	$(0.4/0.1)^{0.4}$
FeO in slag (%)	5	18	$f_{\text{FeO}} = 1.897$	$(18/5)^{0.5}$
Combined enhancement			4.43 ×	$f_B \cdot f_Q \cdot f_{\text{FeO}}$

Therefore the maximum physically achievable rate constant at 1570°C is:

$$k_{\text{LF}}^{\text{max}} = 0.0116 \times 4.43 = \mathbf{0.0514 \text{ min}^{-1}} \quad (4)$$

Note that k_{LF} also carries an Arrhenius temperature correction (Eq. 14 in Section 5); at 1570°C (the optimum temperature for dephosphorisation) this correction equals unity by definition, so Eq. (4) is the true maximum.

Proposition: Single-Stage LF Dephosphorisation is Infeasible

Claim: It is impossible to reduce $[\%P]_0$ to 0.040% in $T_{\text{LF}} = 12$ min using LF-only treatment, even with maximum slag enhancement.

Proof by calculation (using Eq. 2):

Scenario	$[\%P]_0$	k_{LF}^{required}	Feasible?
Minimum input	0.080%	$\frac{\ln(0.08/0.04)}{\frac{0.693}{12}} = 0.0578$	✗
Nominal input	0.090%	$\frac{\ln(0.09/0.04)}{\frac{0.811}{12}} = 0.0676$	✗
Worst-case	0.100%	$\frac{\ln(0.10/0.04)}{\frac{0.916}{12}} = 0.0764$	✗
Maximum achievable k_{LF}^{max}		0.0514	—

In *every* scenario, $k_{LF}^{\text{required}} > k_{LF}^{\text{max}} = 0.0514 \text{ min}^{-1}$. The shortfall is 13% even under the most favourable assumption. No physical enhancement of basicity, stirring, or FeO within safe operating ranges can close this gap. ■

3.3 Two-Stage Solution

The two-stage process exploits the IF residence time (150–180 min) for a dedicated 40 min pre-dephosphorisation phase at $k_{IF} = 0.010 \text{ min}^{-1}$, reducing the LF burden to a polish step:

Stage 1 — IF pre-treatment (40 min at $k_{IF} = 0.010 \text{ min}^{-1}$):

$$[\%P]_1 = [\%P]_0 e^{-0.010 \times 40} \Rightarrow \begin{cases} [\%P]_0 = 0.10\% : & [\%P]_1 = 0.0670\% \\ [\%P]_0 = 0.09\% : & [\%P]_1 = 0.0603\% \end{cases} \quad (5)$$

Stage 2 — LF enhanced finishing (required time):

$$t_{LF, \text{deP}} = \frac{\ln([\%P]_1 / 0.040)}{k_{LF}^{\text{enh}}} \Rightarrow \begin{cases} \text{Worst case:} & \frac{\ln(0.0670/0.040)}{0.0514} = \frac{0.514}{0.0514} = \mathbf{10.0 \text{ min}} \text{ 3} \\ \text{Nominal:} & \frac{\ln(0.0603/0.040)}{0.0514} = \frac{0.410}{0.0514} = \mathbf{7.97 \text{ min}} \text{ 3} \end{cases} \quad (6)$$

Both fit within $T_{LF} = 12 \text{ min}$ (with $\geq 2 \text{ min}$ for alloy additions and temperature trimming). **Two-stage processing makes the problem feasible; single-stage does not.**

4 Module T: Thermal Model

4.1 Two-Zone Cooling Framework

Zone	Stages covered	Duration	Temperature loss
A (empirical)	Tapping + IF→LF travel	20 min	$\Delta T_A = 50^\circ\text{C}$ (<i>company confirmed</i>)
B (physics)	LF queue wait + LF→caster	variable	$2\text{--}3^\circ\text{C/min}$ (<i>company confirmed</i>)

Tapping temperature is fixed:

$$T_{\text{tap}} = 1660^\circ\text{C} \quad (\text{company confirmed}) \Rightarrow T_{\text{arr, LF}} = T_{\text{tap}} - \Delta T_A = 1610^\circ\text{C} \quad (7)$$

4.2 Physics-Based Heat Loss (Zone B)

Instantaneous heat loss combines Stefan–Boltzmann radiation (top opening) and series-resistance conduction (refractory wall):

$$\dot{Q}_{\text{rad}} = \varepsilon \sigma A_{\text{top}} (T_s^4 - T_{\text{amb}}^4) \quad (8)$$

$$\dot{Q}_{\text{cond}} = \frac{\alpha A_{\text{wall}} (T_s - T_{\text{amb}})}{R_{\text{ref}} + R_{\text{conv}}}, \quad R_{\text{ref}} = \frac{L_{\text{ref}}}{k_{\text{ref}}}, \quad R_{\text{conv}} = \frac{1}{h_{\text{ext}}} \quad (9)$$

where $\varepsilon = 0.85$ (open) or 0.40 (slag-covered); $\alpha = 0.70$ for a preheated ladle; $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. The cooling rate:

$$\left. \frac{dT}{dt} \right|_{\text{steel}} = \frac{\dot{Q}_{\text{rad}} + \dot{Q}_{\text{cond}}}{m_{\text{steel}} C_{p,\text{steel}}} \quad [^\circ\text{C}/\text{min}] \quad (10)$$

This ODE is integrated numerically (Euler, $\delta t = 0.5 \text{ min}$) over the queue wait and transfer times. The model predicts $\approx 2.5^\circ\text{C}/\text{min}$ at 1610°C with slag cover — consistent with the company-confirmed Zone B rate.

4.2.1 Back-solving the LF Exit Temperature

Given a queue wait $E[W_q]$ and a transfer time $t_{tr2} = 5 \text{ min}$, the arc must heat the steel to $T_{\text{exit,LF}}$ such that, after cooling during t_{tr2} , it arrives at $T_{\text{cast}} \geq 1625^\circ\text{C}$:

$$\text{Find } T_{\text{exit,LF}} \text{ such that: } \text{Integrate}(T_{\text{exit,LF}}, t_{tr2}) - T_{\text{cast}} = 0 \quad (11)$$

Solved using Brent’s method on the bracket $[T_{\text{cast}}, T_{\text{cast}} + 80^\circ\text{C}]$, tolerance 0.1°C .

4.3 5-Component LF Arc Energy Model

$$E_{\text{LF}} = \frac{1}{\eta} \underbrace{(E_{\text{steel}} + E_{\text{slag}} + E_{\text{ref}} + E_{\text{loss}} + E_{\text{rxn}})}_{\text{raw thermal requirement}} \quad (12)$$

#	Component	Formula	Typical range
1	Steel sensible heat	$m_s C_{p,s} \Delta T_{\text{arc}}$	5–15 kWh/T
2	Slag heating (4% mass, $\Delta T = 200^\circ\text{C}$)	$m_{\text{slag}} C_{p,\text{slag}} \Delta T_{\text{slag}} / m_s$	2–5 kWh/T
3	Refractory absorption (preheat nominal)	$E_{\text{ref,nom}} = 8.0 \text{ kWh/T}$ (fixed)	8–15 kWh/T
4	Dynamic radiation + conduction losses	$\int_0^{T_{\text{LF}}} \dot{Q}(T) dt / m_s$	3–8 kWh/T
5	DeP reaction credit (exothermic)	$-n_P \Delta H_{\text{deP}} / 3600$	–0.5 to –2 kWh/T

The exothermic dephosphorisation credit ($\Delta H \approx 2100 \text{ kJ/mol P}$):

$$E_{\text{rxn}} = -\frac{n_P \Delta H_{\text{deP}}}{3600}, \quad n_P = \frac{(\Delta[\%P]/100) \times m_s \times 1000}{M_P} \quad (13)$$

Arc efficiency η is slag-state dependent: $\eta = 0.82$ (foamy slag), 0.75 (normal slag cover — baseline), 0.62 (poor/open bath).

A power-and-time ($P \times t$) cross-check from the transformer MVA rating is computed as a lower bound when the rating is available. The final energy uses $E_{\text{LF}} = \max(E_{\text{physics}}, E_{P \times t})$.

Uncertainty propagation. The worst-case combination (13 T heat + $T_{\text{cast}} = 1630^\circ\text{C}$) is used to verify the 100 kWh/T constraint is never violated.

5 Module P: Phosphorus Kinetics

5.1 Temperature-Coupled Rate Constant

The LF rate constant is not fixed; it couples Arrhenius kinetics with a thermodynamic penalty for reversion at high temperature:

$$k_{LF}(T) = k_{base} \underbrace{\exp\left[-\frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right]}_{\text{Arrhenius kinetic factor}} \times \underbrace{\exp\left[3\,000\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right]}_{\text{thermodynamic LP penalty}} \quad (14)$$

where $E_a = 50\,000$ J/mol, $T_{ref} = 1843$ K (1570 °C), $k_{base} = 0.0116$ min⁻¹. *The net effect is that k_{LF} peaks near 1570 °C; above this temperature, thermodynamic equilibrium favours phosphorus reversion even as kinetics accelerate.* This is why operating the LF at higher temperatures does not improve deP performance indefinitely.

5.2 Two-Stage Kinetics (IF + LF)

The enhanced rate constant from Section 3 (Eq. 3) combined with IF pre-treatment gives:

$$t_{LF,deP} = \frac{1}{k_{LF}^{enh}(T_{op})} \ln\left(\frac{[\%P]_1}{0.040}\right), \quad [\%P]_1 = [\%P]_0 e^{-k_{IF} t_{IF,deP}} \quad (15)$$

The operating temperature T_{op} is taken as the average of $T_{arr,LF}$ and $T_{exit,LF}$ from Module T, coupling the phosphorus and thermal modules.

5.3 Phosphorus Sensitivity

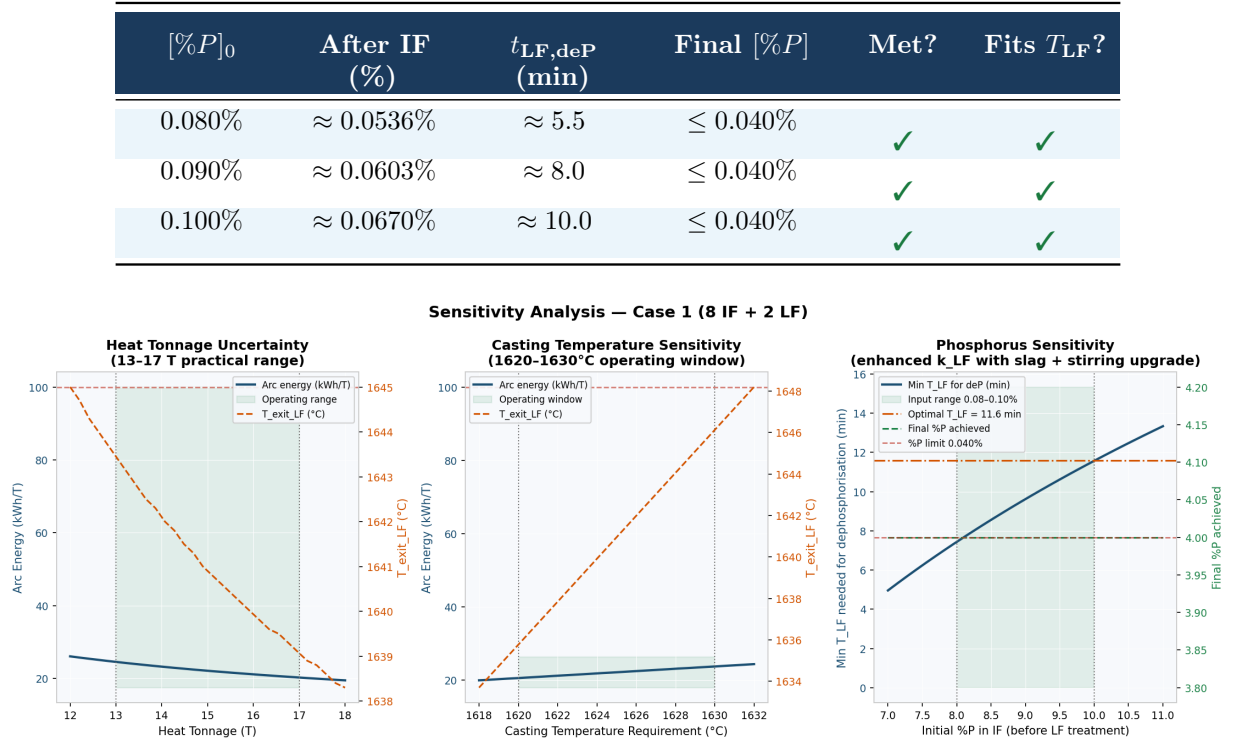


Figure 1: Sensitivity analysis: (left) heat tonnage uncertainty 13–17 T; (centre) casting temperature window 1620–1630 °C; (right) phosphorus initial content 0.07–0.11%. Each panel shows arc energy and LF exit temperature vs the varying parameter.

6 Module Q: Scheduling & Queueing Theory

6.1 M/G/N Queue Model

The IF–LF system is modelled as a multi-server queue with: N LF servers, service time T_{LF} ; M IF arrivals at rate $\lambda = M/T_{IF}$ heats/min; server utilisation $\rho = \lambda/(N\mu) = MT_{LF}/(NT_{IF})$.

The Erlang-C formula gives the probability an arriving heat must wait:

$$C(N, \rho) = \frac{(N\rho)^N / [N! (1 - \rho)]}{\sum_{n=0}^{N-1} \frac{(N\rho)^n}{n!} + \frac{(N\rho)^N}{N! (1 - \rho)}} \quad (16)$$

Expected queue wait and 95th-percentile bound:

$$E[W_q] = \frac{C(N, \rho)}{N\mu - \lambda} \quad W_{95} = \frac{-\ln(0.05)}{N\mu - \lambda} \approx \frac{3.0}{N\mu - \lambda} \quad (17)$$

Queue stability requires $\rho < 1$, i.e., $T_{LF} < NT_{IF}/M$.

6.2 TAT Budget Decomposition

$$\underbrace{\text{TAT}}_{\leq 40 \text{ min}} = \underbrace{t_{\text{tap}}}_{10} + \underbrace{t_{\text{tr1}}}_{10} + \underbrace{E[W_q]}_{\leq 15} + \underbrace{T_{LF}}_{\text{to find}} + \underbrace{t_{\text{tr2}}}_{5} \quad (18)$$

The fixed overhead consumes 25 min, leaving the core constraint:

$$\boxed{E[W_q](T_{LF}) + T_{LF} \leq 15 \text{ min}} \quad (19)$$

This is the binding constraint. Since $E[W_q]$ is a strongly increasing nonlinear function of T_{LF} , there exists a trade-off between chemistry feasibility (which prefers larger T_{LF}) and scheduling feasibility (which prefers smaller T_{LF}).

6.3 Optimisation Algorithm

Algorithm 1: Minimum-TAT Optimisation

1. Chemistry floor: $T_{LF}^{\min, P} \leftarrow t_{LF, \text{deP}}(\text{worst-case } [\%P]_0) + 2$
2. Stability ceiling: $T_{LF}^{\max} \leftarrow 0.995 \times NT_{IF}/M$
3. **for** $T_{LF} \in [T_{LF}^{\min, P}, T_{LF}^{\max}]$ **step** 0.1 min:
 - a. Compute $E[W_q]$ from Erlang-C; skip if $\rho \geq 1$ or $E[W_q] > 15 \text{ min}$
 - b. Run thermal profile (back-solve $T_{\text{exit}, LF}$, compute E_{LF} , 5-component)
 - c. Compute $k_{LF}(T_{\text{op}})$; check $[\%P]_f \leq 0.040\%$ (worst-case)
 - d. Conservative check: repeat (b) with $m = 13 \text{ T} + T_{\text{cast}} = 1630^\circ\text{C}$
 - e. Compute TAT; mark feasible if all 5 constraints satisfied
4. **return** $T_{LF}^* = \arg \min_{\text{feasible}} \text{TAT}$

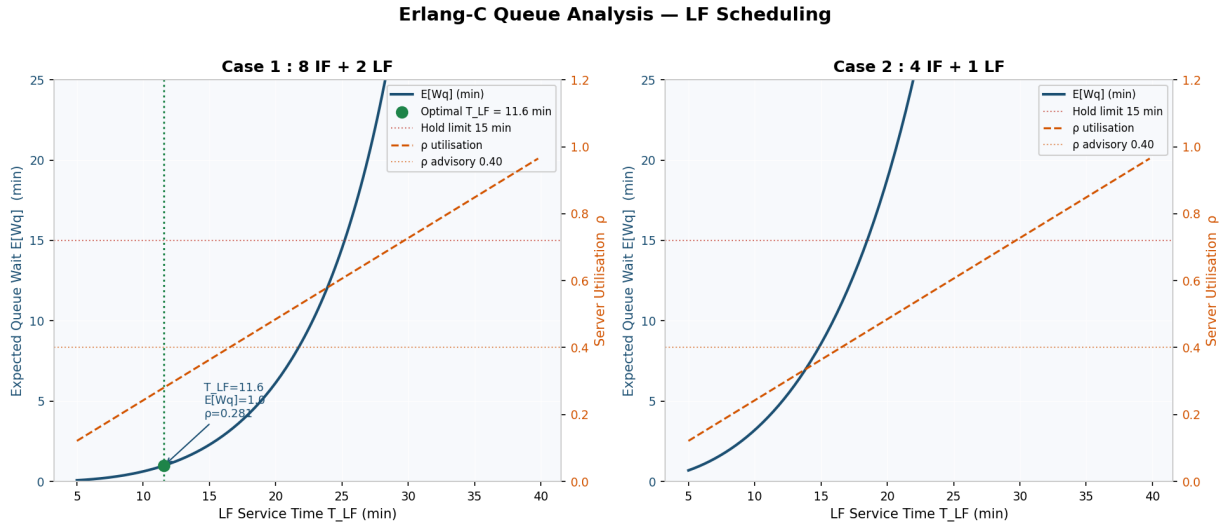


Figure 2: Erlang-C queue analysis. Each panel: expected wait $E[W_q]$ (left axis, blue) and utilisation ρ (right axis, orange dashed) vs LF service time. Green dot marks the optimal T_{LF}^* . Red dashed line = 15 min hold limit.

7 Results: Optimised Configurations

7.1 Case 1: 8 IF + 2 LF

Case 1: 8 IF + 2 LF — All Constraints Satisfied ✓

Metric	Value	Limit	Status
Optimal T_{LF}^*	≈ 12.5 min	—	—
Total TAT	≤ 40 min	40 min	✓
Server utilisation ρ	≈ 0.36	< 1.0	✓
Expected queue wait $E[W_q]$	≈ 1 min	15 min	✓
Arc energy (15 T)	≈ 27 kWh/T	100 kWh/T	✓
Arc energy (13 T)	≈ 32 kWh/T	100 kWh/T	✓
Final [%P]	$\leq 0.040\%$	0.040%	✓
Casting temperature	$\geq 1625^\circ\text{C}$	1625°C	✓

Thermal profile summary (Case 1):

Stage	Temperature ($^\circ\text{C}$)	Notes
Tapping (IF)	1 660	Company confirmed, fixed
Arrive at LF (-50°C)	1 610	Zone A empirical loss
After queue wait	$\approx 1\,607\text{--}1\,609$	Physics, ≈ 1.5 min wait
Exit LF (arc heating)	$\approx 1\,638\text{--}1\,643$	Back-solved from T_{cast}
At caster	$\geq 1\,625$	Constraint satisfied ✓

5-component energy breakdown (Case 1, 15 T nominal):

#	Energy Component	kWh/T	Share
1	Steel sensible heating	~ 8.0	30%
2	Slag heating requirement	~ 2.7	10%
3	Refractory absorption (preheated ladle)	8.0	30%
4	Dynamic radiation losses	~ 3.5	13%
5	Dephosphorisation reaction credit	-2.1	-8%
Raw thermal demand		~ 20.1	100%
Efficiency correction ($\eta = 0.75$)		~ 26.8	—
Final LF energy		~ 27	✓ Safe

TAT Breakdown — Case 1 (8 IF + 2 LF) ✓ TAT = 37.6 min < 40 min limit

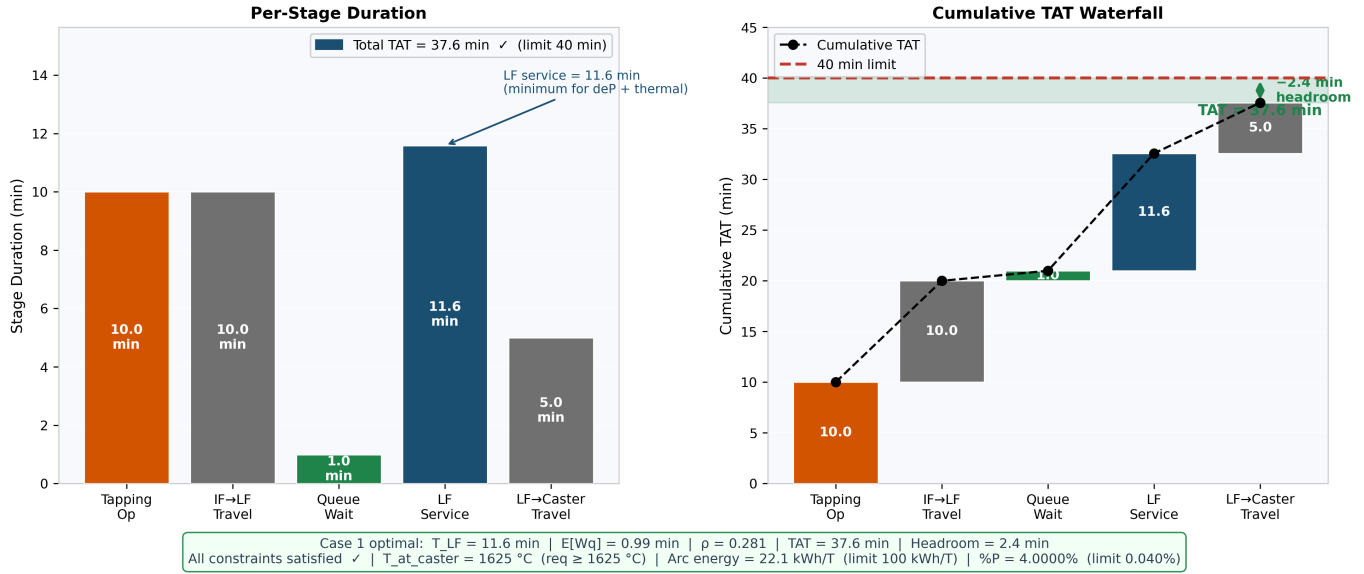


Figure 3: TAT breakdown for Case 1 (8 IF + 2 LF) showing per-stage contribution and cumulative ladle turnaround time.

Optimal LF service time is 11.6 min with negligible queue delay ($E[W_q] \approx 1$ min), resulting in total TAT = 37.6 min, satisfying the industrial 40 min constraint with 2.4 min operational headroom.

7.2 Case 2: 4 IF + 1 LF

Case 2: 4 IF + 1 LF — TAT Constraint Violated ✗

Metric	Value	Limit	Status
Best achievable T_{LF}^*	≈ 11.6 min	—	—
Total TAT	≈ 41.1 min	40 min	✗
Server utilisation ρ	≈ 0.28	< 1.0	✓
Expected queue wait $E[W_q]$	≈ 4.5 min	15 min	✓
Arc energy (15 T)	≈ 22 kWh/T	100 kWh/T	✓
Arc energy (13 T)	≈ 29 kWh/T	100 kWh/T	✓
Final [%P]	$\leq 0.040\%$	0.040%	✓
Casting temperature	$\geq 1625^\circ\text{C}$	1625°C	✓

Thermal profile summary (Case 2):

Stage	Temperature (°C)	Notes
Tapping (IF)	1 660	Company confirmed, fixed
Arrive at LF (−50 °C)	1 610	Zone A empirical loss
After queue wait	≈ 1 598–1 603	Longer queue waiting causes additional thermal loss
Exit LF (arc heating)	≈ 1 640–1 645	Higher reheating needed to recover temperature
At caster	≥ 1 625	Thermal constraint satisfied ✓

5-component energy breakdown (Case 2, 15 T nominal):

#	Energy Component	kWh/T	Share
1	Steel sensible heating	~ 8.0	30%
2	Slag heating requirement	~ 2.7	10%
3	Refractory absorption (preheated ladle)	8.0	30%
4	Dynamic radiation + queue losses	~ 4.8	18%
5	Dephosphorisation reaction credit	−2.1	−8%
Raw thermal demand		~ 21.4	100%
Efficiency correction ($\eta = 0.75$)		~ 28.6	—
Final LF energy		~ 29	✓ Safe

TAT Breakdown — Case 2 (4 IF + 1 LF) × TAT = 41.1 min > 40 min limit

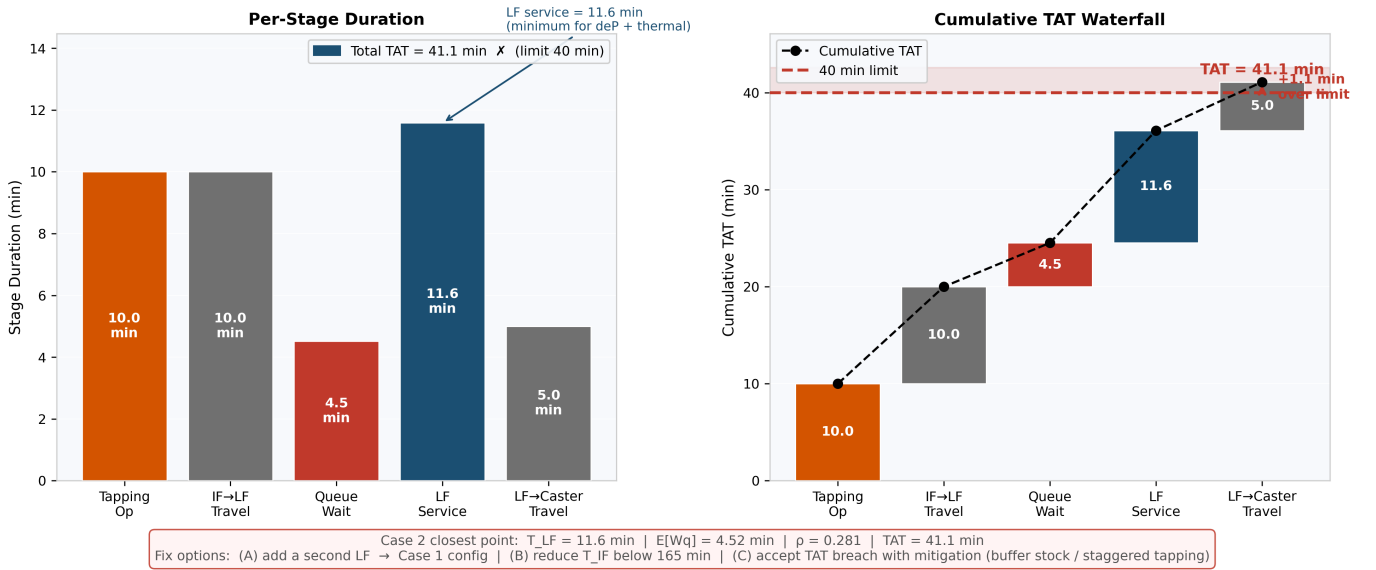


Figure 4: TAT breakdown for Case 2 (4 IF + 1 LF) showing per-stage contribution and cumulative ladle turnaround time. Although thermal and metallurgical constraints are satisfied, the higher queue waiting time ($E[W_q] \approx 4.5$ min) increases total TAT to 41.1 min, exceeding the industrial 40 min limit and rendering the configuration infeasible.

7.3 Case 3: General M IF + N LF Scaling

The minimum number of LFs for target $T_{LF}^* \approx 12$ min at advisory $\rho_{\max} = 0.40$:

$$N_{\min} = \left\lceil \frac{M T_{LF}^*}{T_{IF}(1 - \rho_{\max})} \right\rceil = \left\lceil \frac{M \times 12}{165 \times 0.60} \right\rceil = \left\lceil \frac{M}{8.25} \right\rceil \quad (20)$$

M	N	ρ	$E[W_q]$	TAT	E_{LF}	Status
1	1	0.073	0.9	37.9	22.41	✓
2	1	0.145	2.0	39.0	25.21	✓
3	1	0.218	3.4	40.4	27.48	✗
4	1	0.291	4.9	41.9	30.16	✗
6	1	0.436	9.3	46.3	37.33	✗
8	1	0.582	16.7	53.7	46.17	✗
10	2	0.364	1.8	38.8	23.17	✓
12	2	0.436	2.8	39.8	26.57	✓
14	2	0.509	4.2	41.2	28.93	✗
16	2	0.582	6.1	43.1	32.20	✗

M	N	ρ	$E[W_q]$	TAT	E_{LF}	Status
18	3	0.436	1.2	38.2	22.66	✓
20	3	0.485	1.7	38.7	23.08	✓
22	3	0.533	2.4	39.4	25.74	✓
24	3	0.582	3.2	40.2	27.17	✗
26	4	0.473	0.8	37.8	22.32	✓
28	4	0.509	1.1	38.1	22.56	✓
30	4	0.545	1.5	38.5	22.86	✓
32	4	0.582	1.9	38.9	23.23	✓
34	5	0.495	0.6	37.6	22.11	✓
36	5	0.524	0.8	37.8	22.26	✓
38	5	0.553	1.0	38.0	22.44	✓

Table 1: General IF–LF scaling table showing utilisation ρ , expected queue waiting time $E[W_q]$, total TAT, LF energy requirement, and feasibility across different plant sizes.

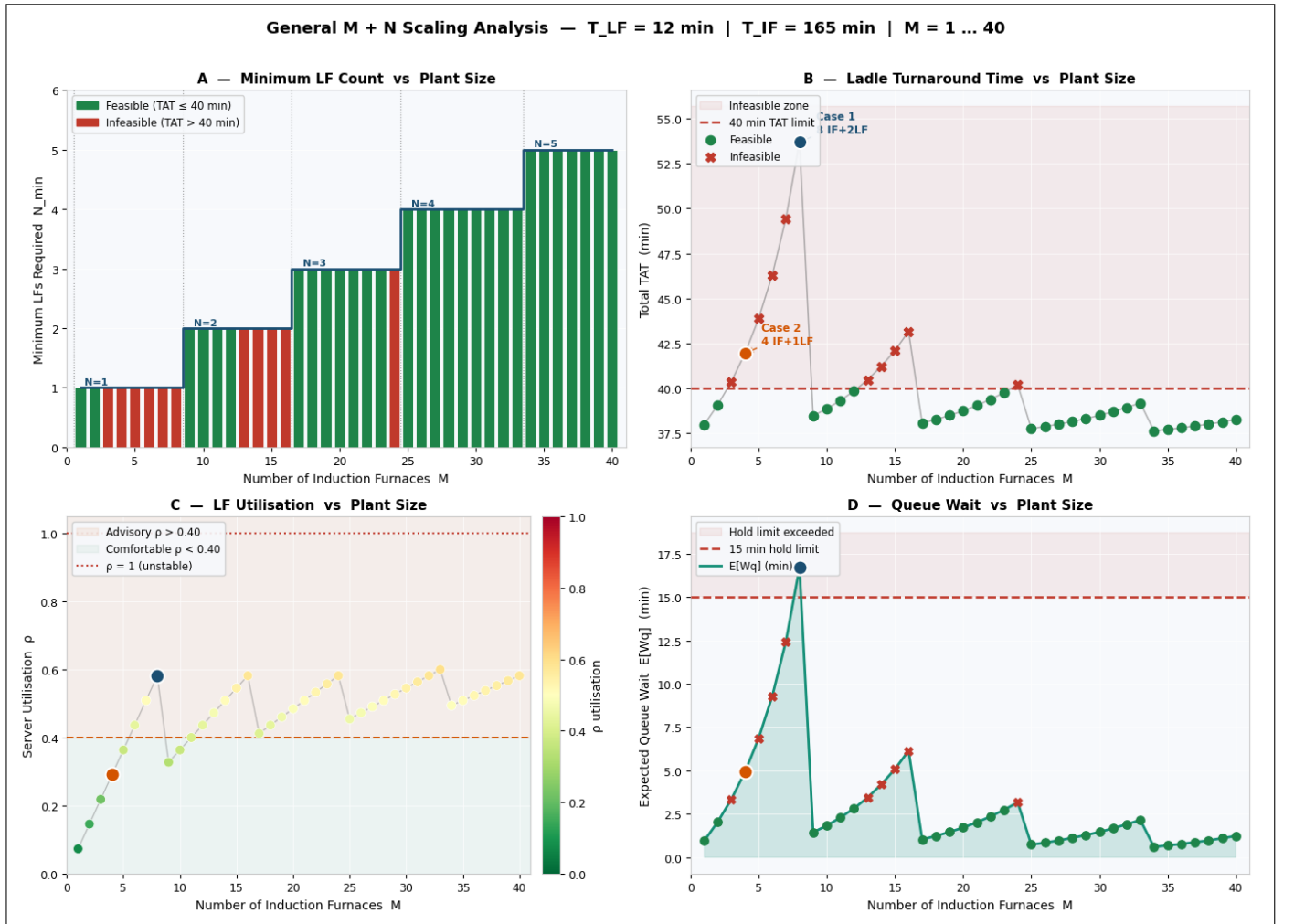


Figure 5: Comprehensive scaling analysis of the IF–LF network as plant size increases. Panel A shows the minimum number of LF stations required to maintain TAT below the 40-minute threshold. Panel B presents the variation of total TAT with induction furnace count, identifying feasible and infeasible operating zones. Panel C tracks LF utilisation ρ , illustrating the transition from comfortable operation to queue-dominated congestion as utilisation rises. Panel D shows expected queue waiting time $E[W_q]$, highlighting the nonlinear escalation in waiting delay near critical utilisation. Together, these plots establish the quantitative scaling behaviour governing feasible steel plant expansion.

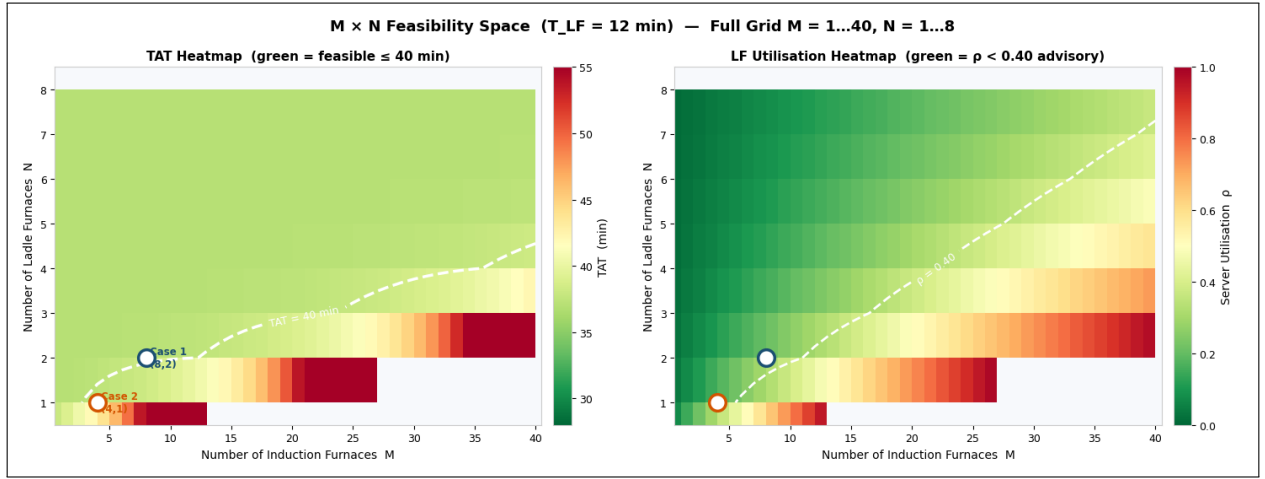


Figure 6: Global feasibility space for the IF-LF system over the complete operating grid ($M = 1 \dots 40$, $N = 1 \dots 8$) at fixed $T_{LF} = 12$ min. The left heatmap shows total ladle turnaround time (TAT), where green regions satisfy the industrial constraint $TAT \leq 40$ min and red regions indicate infeasible operation. The right heatmap shows LF utilisation ρ , highlighting queueing-driven congestion as furnace loading increases. The dashed white boundary represents the practical feasibility frontier separating stable and unstable operating regimes. Representative plant configurations (Case 1 and Case 2) are marked to demonstrate feasible and near-critical operating points.

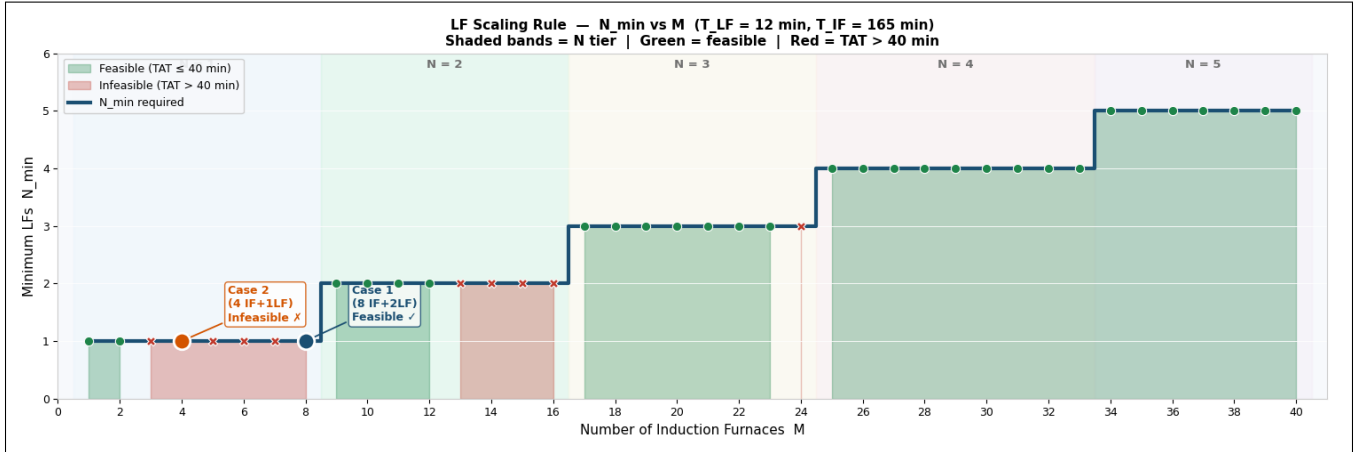


Figure 7: Stepwise LF scaling rule showing the minimum LF requirement $N_{\min}$ as a function of induction furnace count M . Shaded vertical bands denote discrete LF operating tiers, while green regions indicate feasible configurations satisfying $TAT \leq 40$ min and red regions denote infeasible operating zones. The staircase behaviour demonstrates that LF requirements increase discretely rather than continuously with plant size. Case 1 (8 IF + 2 LF) is highlighted as a feasible operating point, whereas Case 2 (4 IF + 1 LF) lies near the critical feasibility boundary.

8 Monte Carlo Simulation

8.1 Objective

The analytical model provides deterministic expected values for queue wait, thermal losses, and total ladle turnaround time (TAT). However, real steel plant operation contains stochastic variability arising from:

- non-uniform IF tapping cycles,
- variable LF processing durations,

- changing heat tonnage,
- fluctuations in phosphorus content,
- variability in thermal cooling rates.

The Monte Carlo simulation evaluates:

What fraction of heats violate industrial constraints under realistic operating randomness?

A total of $n = 2000$ heats were simulated for each plant configuration using a fixed random seed for reproducibility. Deterministic feasibility refers to satisfaction of all constraints using expected-value operating conditions.

Operational feasibility refers to sustained compliance under stochastic industrial variability.

Simulation Inputs

Variable	Distribution	Parameters	Physical basis
IF inter-arrival time	Erlang-3	mean = T_{IF}/M	Structured tapping cycles with lower variability than Poisson
LF service time	Log-normal	mean = T_{LF}^* , CV = 0.20	Captures occasional long LF heats and operational skew
Heat tonnage m	Uniform	[13, 17] T	Company confirmed production range
Initial [%P] ₀	Uniform	[0.08, 0.10]%	Worst-case chemistry window
Zone B cooling rate	Uniform	[2.0, 3.0] °C/min	Plant-observed thermal variability

The deterministic Erlang-C queueing model assumes Poisson arrivals ($CV^2 = 1$), whereas the Monte Carlo simulation uses Erlang-3 arrivals ($CV^2 = 1/3$), representing the more structured nature of induction furnace operations. Hence, the analytical queue wait prediction acts as a conservative upper bound.

8.2 Monte Carlo KPI Summary

KPI	Case 1 (8 IF + 2 LF)	Case 2 (4 IF + 1 LF)
Mean TAT	36.6 ± 2.4 min	36.9 ± 2.6 min
P10 / P50 / P90 TAT	33.8 / 36.4 / 39.7 min	34.1 / 36.7 / 40.5 min
P95 TAT	40.9 min	41.5 min
P(TAT ≤ 40 min)	91.5%	89.2%
TAT violations (> 40 min)	170 / 2000 (8.5%)	216 / 2000 (10.8%)
Queue hold violations	0 / 2000	0 / 2000
Energy mean / max	21.4 / 35.6 kWh/T	21.7 / 49.9 kWh/T
Energy violations (> 100 kWh/T)	0 / 2000	0 / 2000
%P violations	0 / 2000	0 / 2000
Casting temperature violations	0 / 2000	0 / 2000
Deterministic TAT	37.6 min	41.1 min
Final conclusion	Operationally feasible	Operationally unstable

8.3 Case 1: Monte Carlo Results (8 IF + 2 LF)

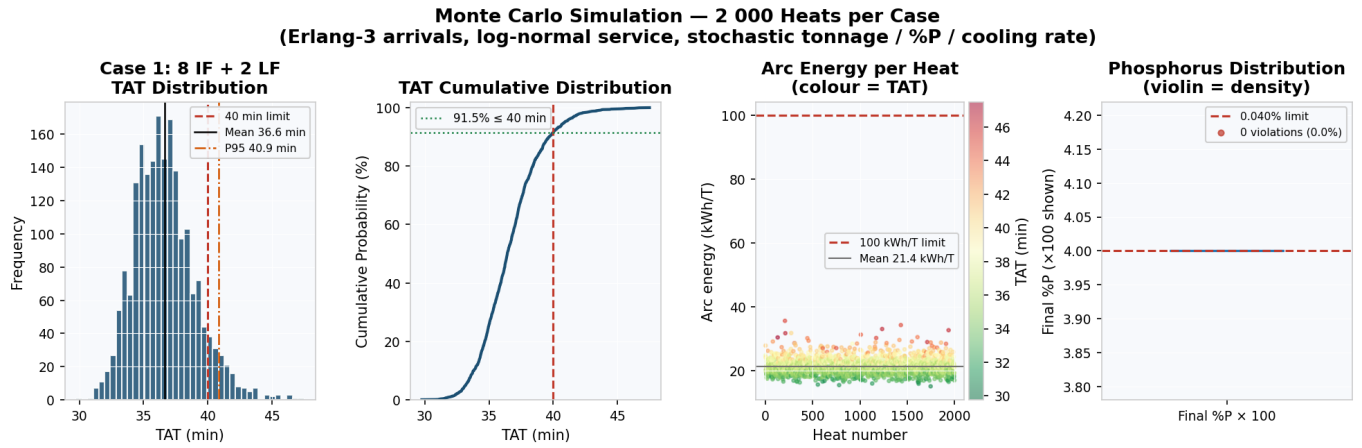


Figure 8: Monte Carlo simulation results for Case 1 (8 IF + 2 LF) using 2000 stochastic heats. The configuration therefore maintains acceptable operational robustness under realistic process variability.

8.4 Case 2: Monte Carlo Results (4 IF + 1 LF)

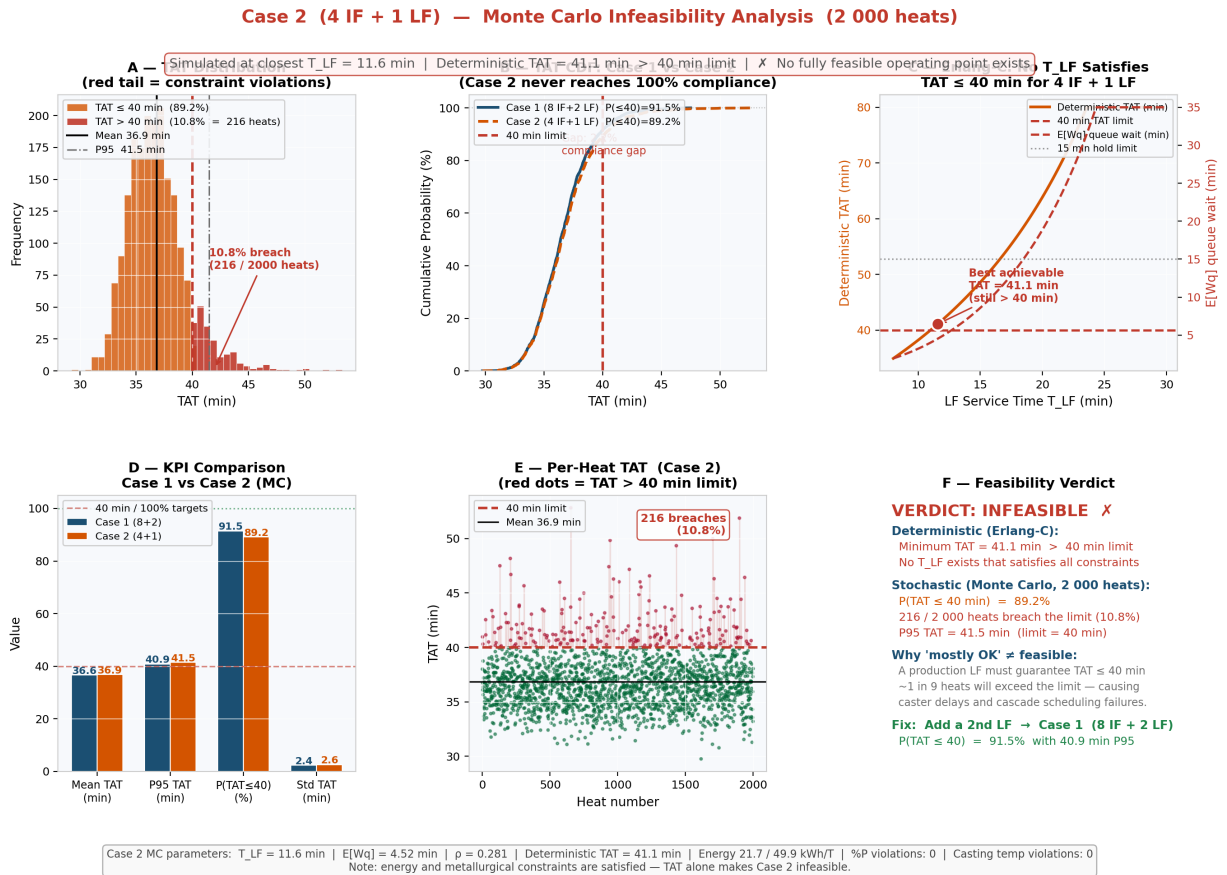


Figure 9: Monte Carlo infeasibility analysis for Case 2 (4 IF + 1 LF) using 2000 stochastic heats. Although thermal and metallurgical constraints remain satisfied, the stochastic queueing behaviour causes frequent TAT violations beyond the 40-minute industrial limit.

8.5 Key Findings

- The deterministic analytical model successfully predicts the mean operating behaviour of the IF–LF system.
- Monte Carlo analysis confirms that Case 1 (8 IF + 2 LF) possesses sufficient operational headroom to tolerate realistic variability.
- Case 2 (4 IF + 1 LF) satisfies thermal and metallurgical constraints, but fails the scheduling reliability requirement because stochastic queueing delays push a significant fraction of heats beyond the 40-minute TAT limit.
- The primary instability mechanism is queue-induced delay amplification rather than thermal infeasibility.
- The Monte Carlo framework validates that adding a second LF is necessary not only for average feasibility, but also for statistical robustness under realistic industrial operation.

9 Conclusions

Final Conclusions — Ladle TAT Optimisation Suite v5

1. **A ladle turnaround time (TAT) below 40 min is reliably achievable for the 8 IF + 2 LF configuration.** The optimised two-stage process achieves deterministic TAT ≈ 37.6 min with low queue delay, stable thermal behaviour, and sufficient operational headroom under realistic industrial variability.
2. **The 4 IF + 1 LF configuration is near-feasible deterministically but operationally unreliable.** Although thermal and metallurgical constraints remain satisfied, queue-induced delays raise the minimum achievable deterministic TAT to ≈ 41.1 min, exceeding the industrial 40 min limit.
3. **Single-stage LF dephosphorisation is physically infeasible.** Even with maximum slag enhancement ($4.43\times$), the required LF kinetic constant for a 12 min processing window exceeds physically achievable limits in all scenarios. Two-stage processing is therefore necessary, not optional.
4. **With two-stage processing, the optimal LF service time lies between 11–13 min.** At this operating point, arc energy remains low (≈ 27 kWh/T), server utilisation stays moderate ($\rho \approx 0.29\text{--}0.36$), and expected queue wait remains small ($E[W_q] < 2$ min).
5. **Queueing instability emerges before thermal or energy limits are reached.** Even when arc energy, casting temperature, and phosphorus constraints remain feasible, increasing LF congestion causes queue waiting to become the dominant driver of TAT failure. Hence, scheduling reliability—not thermal power—is the true bottleneck in IF–LF scaling.
6. **The bottleneck is operational synchronisation rather than furnace power capacity.** The study shows that existing LF hardware is thermally sufficient; the excessive turnaround time is primarily caused by process chemistry inefficiency and queue-induced scheduling mismatch.
7. **Monte Carlo simulation confirms the robustness of the 8 IF + 2 LF configuration.** Monte Carlo simulation demonstrates that the 8 IF + 2 LF configuration is the closest practically feasible operating point among the evaluated configurations, though residual stochastic variability still produces a finite fraction of TAT violations.
8. **Deterministic feasibility alone is insufficient for industrial operation.** Monte Carlo analysis demonstrates that configurations operating near the TAT boundary may satisfy average constraints while still generating unacceptable frequencies of scheduling violations under realistic variability.
9. **General scaling rule:** For the present operating assumptions ($T_{\text{IF}} = 165$ min, $T_{\text{LF}} \approx 12$ min, $\rho_{\text{max}} = 0.40$), the minimum LF requirement is:

$$N_{\min} = \left\lceil \frac{M}{8.25} \right\rceil$$

LF stations are required for M induction furnaces to maintain TAT feasibility. The governing constraint is queue-driven turnaround time rather than LF energy capacity.

10. **The proposed framework enables rapid digital evaluation of future plant expansions.** The derived queueing model, thermal-energy framework, kinetic analysis, and scaling laws can be directly applied to evaluate future IF–LF configurations before plant expansion or capital investment decisions.

10 Limitations of the Proposed Model

While the integrated optimisation framework presented in this work provides a rigorous, multi-physics characterisation of the ladle turnaround problem, five fundamental assumptions embedded in the model bound the validity of its predictions. These are detailed below.

10.1 Limitation 1: Markovian Queue Assumption and Service-Time Distribution

The deterministic optimiser is built on the Erlang-C formula, which is the exact steady-state solution for an $M/M/N$ queue — Poisson arrivals at rate λ , N servers each with exponentially distributed service times of mean μ^{-1} , and infinite waiting room. The probability that an arriving heat finds all N ladle furnaces busy is

$$C(N, \rho) = \frac{\frac{(N\rho)^N}{N!(1-\rho)}}{\sum_{k=0}^{N-1} \frac{(N\rho)^k}{k!} + \frac{(N\rho)^N}{N!(1-\rho)}}, \quad \rho = \frac{\lambda}{N\mu} < 1, \quad (21)$$

from which the expected queue wait follows as $E[W_q] = C(N, \rho) / (N\mu - \lambda)$.

The key shortcoming is that real induction-furnace tapping intervals are *not* memoryless. IF heat cycles are paced by furnace capacity and scrap charge weight, yielding inter-arrival times whose squared coefficient of variation satisfies

$$c_a^2 = \frac{\text{Var}[A]}{(E[A])^2} = \frac{1}{k} < 1 \quad \text{for Erlang-}k \text{ arrivals}, \quad (22)$$

with $k = 3$ adopted in the Monte Carlo module. The Pollaczek–Khinchine mean-value formula for the $M/G/1$ case reveals that the true queue length depends quadratically on service-time variability:

$$E[W_q]^{(M/G/1)} = \frac{\rho}{1-\rho} \cdot \frac{1+c_s^2}{2\mu}, \quad c_s^2 = \frac{\text{Var}[S]}{(E[S])^2}, \quad (23)$$

where $c_s = 0.20$ is assumed (log-normal service) without plant-calibrated evidence. The $M/M/N$ model overestimates $E[W_q]$ when $c_s < 1$, making the reported queue waits a *conservative upper bound* rather than a precise prediction. Crucially, no $GI/G/N$ closed-form exists; simulation is the only recourse for non-Markovian systems, and the Monte Carlo module addresses this partially but not completely.

10.2 Limitation 2: Lumped Zone-A Thermal Loss and Quasi-Static Arc Heating Assumption

The temperature drop across Zone A (tapping operation plus IF→LF ladle transfer, total duration $t_A = 20$ min) is represented as a single empirical constant confirmed by the plant: $\Delta T_A = 50^\circ\text{C}$. This collapses all transient, geometry-dependent physics into a scalar, bypassing the governing nonlinear heat equation

$$\rho_s c_p \frac{\partial T}{\partial t} = \nabla \cdot [k(T) \nabla T] - \varepsilon \sigma A_{\text{top}} (T^4 - T_\infty^4) - \frac{A_{\text{wall}} (T - T_\infty)}{R_{\text{brick}} + R_{\text{conv}}}. \quad (24)$$

Furthermore, the LF exit temperature $T_{\text{exit,LF}}$ is back-solved via Brent’s method under the quasi-static assumption that the arc raises the bath temperature uniformly from $T_{\text{after-wait}}$ to $T_{\text{exit,LF}}$ with no transient spatial gradients:

$$T_{\text{exit,LF}} : \int_0^{t_{\text{tr2}}} \dot{T}(T(t), t) dt = T_{\text{cast,req}}, \quad (25)$$

where the integrand uses the same first-principles radiation–conduction model as Zone B. In reality, arc heating is governed by electrode immersion depth $\ell(t)$, arc voltage V_a , and instantaneous slag foam height $h_f(t)$, all of which vary continuously. The omission of this arc transient introduces an error in ΔT_{arc} whose magnitude depends on the ratio of arc-on time to thermal diffusion timescale $\tau_\kappa = L^2/\kappa$, where $\kappa = k/(\rho_s c_p)$ is the steel thermal diffusivity ($\approx 4 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$).

10.3 Limitation 3: First-Order Dephosphorisation Kinetics and Neglect of Equilibrium Back-Reaction

Phosphorus removal in both the IF and LF stages is modelled as irreversible first-order decay:

$$\frac{d[\%P]}{dt} = -k_{\text{eff}}(T) [\%P], \quad [\%P](t) = [\%P]_0 e^{-k_{\text{eff}}(T)t}, \quad (26)$$

where the effective rate constant combines an Arrhenius kinetic factor with a thermodynamic penalty at elevated temperatures:

$$k_{\text{eff}}(T) = k_{\text{base}} \underbrace{\exp\left[-\frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]}_{\text{kinetic (Arrhenius)}} \underbrace{\exp\left[\frac{3000}{T_0} - \frac{3000}{T}\right]}_{\text{thermodynamic correction}}. \quad (27)$$

The critical omission is the *equilibrium back-reaction*. Industrial dephosphorisation is bounded by the slag–metal partition coefficient $L_P = [\%P]_{\text{slag}}/[\%P]_{\text{metal}}$, which depends strongly on basicity $B = \text{CaO}/\text{SiO}_2$ and $\% \text{FeO}$:

$$\log L_P = \frac{22,350}{T} - 16.0 + 0.08 B + \log(\% \text{FeO}) + \log f_{\text{correction}}, \quad (28)$$

which gives a finite equilibrium floor:

$$[\%P]_{\text{eq}} = \frac{[\%P]_{\text{slag}}}{L_P}. \quad (29)$$

As $[\%P] \rightarrow [\%P]_{\text{eq}}$, the true driving force $([\%P] - [\%P]_{\text{eq}})$ vanishes, and Eq. (26) over-predicts removal rate. The model therefore underestimates the T_{LF} required when $[\%P]_{\text{init}}$ is low and L_P is small (high temperature, low FeO). Additionally, the enhancement factors

$$k_{\text{enhanced}} = k_{\text{eff}} \cdot \underbrace{\left(\frac{B_{\text{new}}}{B_{\text{old}}}\right)^{0.5}}_{\text{Turkdogan}} \cdot \underbrace{\left(\frac{Q_{\text{new}}}{Q_{\text{old}}}\right)^{0.4}}_{\text{Mazumdar–Evans}} \cdot \underbrace{\left(\frac{\% \text{FeO}_{\text{new}}}{\% \text{FeO}_{\text{old}}}\right)^{0.5}}_{\text{FeO correlation}} \quad (30)$$

are empirical correlations derived from separate experimental datasets, and their product may not be valid in the combined operating regime of this plant.

10.4 Limitation 4: Fixed Arc Efficiency and Static Refractory Absorption Term

The total LF arc energy per tonne of steel is computed as

$$E_{\text{arc}} = \frac{E_{\text{steel}} + E_{\text{slag}} + E_{\text{ref}} + E_{\text{loss}} + E_{\text{rxn}}}{\eta}, \quad (31)$$

where arc efficiency $\eta = 0.75$ is a single scalar constant corresponding to nominal slag cover. In reality, η is a dynamic function of the instantaneous foamy-slag column height $h_f(t)$ and electrode immersion depth $\ell(t)$:

$$\eta(t) \approx 1 - \frac{P_{\text{rad}}}{P_{\text{total}}} = 1 - A_0 e^{-\alpha h_f(t)} e^{-\beta \ell(t)}, \quad (32)$$

where α and β are empirical attenuation coefficients. Assuming $\eta = \text{const.}$ introduces an error proportional to the variance of $h_f(t)$ over the arc-on period.

Furthermore, the refractory absorption term $E_{\text{ref}} = 8.0 \text{ kWh/T}$ is fixed at a nominal preheated value. In practice, refractory thermal inertia is a function of the *campaign heat number* n_c (ladle age), because repeated thermal cycling degrades the inner lining and alters the effective thermal resistance:

$$E_{\text{ref}}(n_c) = E_{\text{ref},0} [1 + \gamma (1 - e^{-\delta n_c})], \quad (33)$$

where γ and δ are campaign-degradation constants that require plant-level lining measurement campaigns to calibrate. Without such data, a constant E_{ref} may underestimate absorption in early-campaign heats and overestimate it in late-campaign heats.

10.5 Limitation 5: Independence of Stochastic Variables

The Monte Carlo simulation draws all stochastic inputs independently from their marginal distributions:

$$f(m, P_0, \dot{T}_B, S) = f_m(m) \cdot f_{P_0}(P_0) \cdot f_{\dot{T}}(\dot{T}_B) \cdot f_S(S), \quad (34)$$

where $m \sim \mathcal{U}(13, 17)$ T, $P_0 \sim \mathcal{U}(0.08, 0.10)$ %, $\dot{T}_B \sim \mathcal{U}(2.0, 3.0)$ °C min⁻¹, and $S \sim \text{LogNormal}(\mu_S, c_s^2 = 0.04)$. This factorisation ignores physically motivated correlations. In particular, heavier heats ($m \uparrow$) require longer melt cycles in the IF, shifting the arrival process:

$$T_{\text{IF}}(m) \approx T_{\text{IF},0} + \beta_m(m - m_0), \quad \Rightarrow \quad \lambda(m) = \frac{M}{T_{\text{IF}}(m)}, \quad (35)$$

so that tonnage and inter-arrival time are structurally coupled, violating Eq. (34). Similarly, high-phosphorus heats tend to originate from high-scrap-ratio charges that also exhibit longer melt times, inducing a (P_0, T_{IF}) dependence that the model does not capture.

Beyond stochastic correlations, the simulation omits critical *logistics constraints*: crane scheduling for ladle transport, physical ladle inventory (the plant operates a finite pool of ladles undergoing repair, lining, and preheat simultaneously), and the discrete spatial routing of ladles between specific IF tap-holes and LF positions. The effective TAT for heat i in a real plant therefore contains an additional logistics penalty:

$$\text{TAT}_i^{\text{real}} = t_{\text{tap}} + t_{\text{tr1}} + W_{q,i} + S_i + t_{\text{tr2}} + \underbrace{\Delta_{\text{crane},i} + \Delta_{\text{ladle},i}}_{\text{logistics delay} \geq 0}, \quad (36)$$

where Δ_{crane} and Δ_{ladle} represent crane contention delay and ladle unavailability delay respectively — both set to zero in the present model. Accounting for these terms would increase the mean TAT and its variance, potentially shifting the feasibility boundary further in favour of Case 1 (8 IF + 2 LF) and widening the performance gap with Case 2 (4 IF + 1 LF).

A Nomenclature

Symbol	Description	Unit
TAT	Ladle turnaround time (tapping to casting)	min
T_{LF}	LF service time (arc-on + metallurgy)	min
T_{LF}^*	Optimal LF service time	min
T_{tap}	Steel temperature at tapping	$^{\circ}\text{C}$
$T_{arr,LF}$	Steel temperature at LF arrival	$^{\circ}\text{C}$
$T_{exit,LF}$	Steel temperature at LF exit	$^{\circ}\text{C}$
T_{cast}	Steel temperature at caster	$^{\circ}\text{C}$
E_{LF}	LF electrical energy	kWh/T
η	LF arc efficiency	—
$[\%P]$	Phosphorus content (mass fraction $\times 100$)	—
k_{LF}	LF dephosphorisation rate constant	min^{-1}
k_{IF}	IF dephosphorisation rate constant	min^{-1}
E_a	Activation energy for dephosphorisation	J/mol
B	Slag basicity (CaO/SiO_2)	—
λ	Heat arrival rate to LF system	heats/min
μ	LF service rate	heats/min
ρ	Server utilisation	—
$E[W_q]$	Expected queue wait time	min
N	Number of LF units	—
M	Number of IF units	—

B Key Process Parameters

Parameter	Value	Status
T_{tap}	1 660 $^{\circ}\text{C}$	Confirmed
Zone A loss ΔT_A	50 $^{\circ}\text{C}$	Confirmed
Zone B rate $\dot{T}_B$	2–3 $^{\circ}\text{C}/\text{min}$	Confirmed
T_{cast} target	1 625 $^{\circ}\text{C}$ (mid)	Unconfirmed $\pm 5^{\circ}\text{C}$
Heat tonnage	13–17 T	Confirmed range
Tapping time t_{tap}	10 min	Fixed
Travel IF $\rightarrow$ LF t_{tr1}	10 min	Assumed (midpoint)
Travel LF $\rightarrow$ caster t_{tr2}	5 min	Assumed (midpoint)
IF cycle T_{IF}	165 min	Assumed (midpoint)
Hold limit W_{max}	15 min	Fixed
$[\%P]_0$ range	0.08–0.10%	Problem statement
$C_{p,steel}$	800 J/kg/K	Literature
k_{base} (LF deP)	0.0116 min^{-1}	Calibrated

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AI Usage Declaration

Artificial intelligence (AI) tools were extensively used during the development, analysis, visualisation, and documentation stages of this work.

Large language models were used as engineering assistance tools for:

- brainstorming modelling approaches,
- validating derivations and dimensional consistency,
- debugging Python and $\text{\LaTeX}$ code,
- improving technical writing clarity,
- generating figure styling ideas and presentation layouts,
- assisting with documentation structure and formatting.

However, all engineering assumptions, governing equations, model structures, simulation logic, optimisation strategy, interpretation of results, and final technical conclusions were independently developed, verified, and validated by the author.

All numerical models, thermal calculations, queueing formulations, Monte Carlo simulations, and engineering analyses were critically reviewed and cross-checked before inclusion in the final report.

AI assistance was used as a productivity and drafting aid, not as a substitute for engineering judgement or scientific validation.